# Seminar International Financial Markets

## Week 2

Dr. Yao Chen

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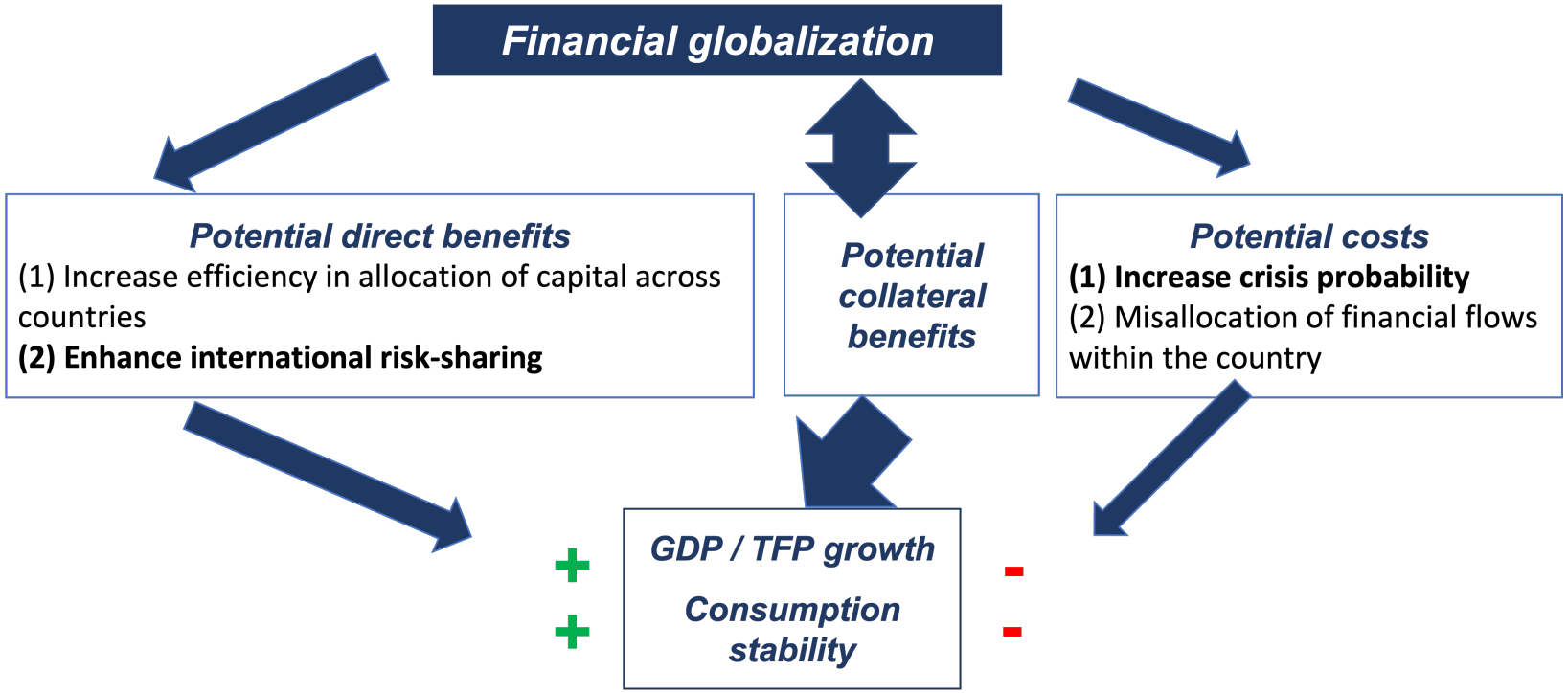
# Recap: theoretical costs and benefits of financial globalization



# Outline

**International risk sharing**

**Sovereign default**

# International risk sharing

A simple model of international asset trade

Optimal risk sharing

Market incompleteness and risk sharing

# International risk sharing and consumption smoothing

- **Risk aversion** $\rightarrow$ for a given level of average consumption, stable consumption is preferred
- If no trading in financial assets (ignoring capital investment): income $=$ consumption. Thus, volatile income $\rightarrow$ volatile consumption
- A world capital market allows countries with **stochastic** income to share **risk** and stabilize consumption

  $\rightarrow$ a **theoretical benefit** from participating in the international financial markets

# A simple model of international asset trade with output risk

Two periods:

- Period 1: The country has a fixed and perishable output of $y_1 = \bar{y}$
- Period 2: Exogenous output depends on "states of nature" $s$

$$y_2 = y_2(s), \quad s \in S, \quad prob(s) = \pi(s), \qquad \sum_{s \in S} \pi(s) = 1$$

# States of nature

- In model with **uncertain** situation, any one of several possible outcomes may be realized in each period
- Each **state** is an **exhaustive** description of **exactly one possible outcome**
- The set of **all states of nature** $S$ includes everything that can happen

An example:

| | States of nature | |
|---|---|---|
| Output | more rain | less rain |
| Country 1 produces sunflower | 0.5 | 1 |
| Country 2 produces rice | 1 | 0.5 |

In our model, each **state of nature** is associated with a specific level of output

# A simple model of international asset trade with output risk (cont.)

- 2 countries, Home and Foreign (denoted by $*$): each populated by a representative household
- Prices of output are identical across countries (law of one price)
- Common "states of nature"
- Risk aversion
- Households' lifetime utility:

$$U = u(c_1) + \beta \sum_s \pi(s) u[c_2(s)], \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma}$$

where the time discount factor is $\beta \in (0, 1)$, $\sigma > 0$ reflects risk aversion, and $c_t(s)$ is the consumption in period $t$ and state $s$

# Motivation: sharing risk across states

Back to the previous example

| | States of nature | |
|---|---|---|
| Output | more rain | less rain |
| Country 1 produces sunflower | 0.5 | 1 |
| Country 2 produces rice | 1 | 0.5 |

**Discussion**

- What kind of asset would allow countries to **insure against rainfall risk**?

# Arrow-Debreu securities (AD)

- One-period **state-contingent claims**:
  pay 1 unit of output **if and only if** a particular state of nature $s$ realizes at a particular date
- AD securities are traded on competitive markets in period 1. An $AD(s)$ security is traded at the price $q(s)$ (in units of period 1 consumption)
- Financial contracts are always enforced

# Arrow-Debreu securities: an example

$s \in \{\text{more rain, less rain}\}$

| | States of nature | |
|---|---|---|
| Payment in t=2 | more rain | less rain |
| AD(more rain) | 1 | 0 |
| AD(less rain) | 0 | 1 |

# Market completeness

- Markets are **complete** if there exists an AD for each state of nature $s \in S$
- we can replicate **every** financial asset as a portfolio of ADs $\Rightarrow$ any other financial market is redundant

**Why do we care about complete financial markets?**

Under complete financial markets, market equilibrium = **first-best allocation**

After finding the first-best allocation, we then have a benchmark to compare with the actual situation

# Market completeness: an example

**Market completeness**: there exists an AD for each state of nature

| | States of nature | |
|---|---|---|
| Payment in t=2 | more rain | less rain |
| AD(more rain) | 1 | 0 |
| AD(less rain) | 0 | 1 |
| **Risk-free asset** | 1 | 1 |
| **Replicating the risk-free asset with ADs** | | |
| 1 AD(more rain) +1 AD(less rain) | 1 | 1 |
| **A generic asset** | X | Y |
| **Replicating the generic asset with ADs** | | |
| X*AD(more rain) + Y*AD(less rain) | X | Y |

# Market completeness (cont.)

- Markets are **complete** if there exists an AD for each state of nature $s \in S$
- **No arbitrage condition**: the price of an asset = the sum of prices of ADs that replicate the payment of the asset

  For a risk-free asset:

$$\sum_s q(s) = \frac{1}{R}, \quad R : \text{risk-free interest rate}$$

- Markets are **incomplete** if there are not enough independent AD to cover all states of nature

# Optimal risk sharing under complete financial markets

The representative household's maximization problem, given $q(s)$

$$\max_{c_1, \{b(s), c_2(s)\}_{s=1,\ldots,S}} \quad u(c_1) + \beta \sum_s \pi(s) u[c_2(s)] \tag{1}$$

$$\text{subject to} \quad c_1 = y_1 - \sum_s q(s)\, b(s)$$

$$c_2(s) = y_2(s) + b(s), \quad \forall \mathbf{s} \in \mathbf{S}$$

$b(s)$: AD for the state $s$

Notice that in period 2 we have **1 budget constraint for each state**

Notice also the intertemporal budget constraint:

$$c_1 + \sum_s q(s) c_2(s) = y_1 + \sum_s q(s) y_2(s)$$

Complete international financial markets allow the country to allocate consumption **across time**, i.e., $t = 1, 2$, and **across states of nature**, i.e., $s \in S$

# Optimal risk sharing under complete markets (cont.)

Set up a Lagrangian for the household's maximization problem

$$\begin{aligned}\mathcal{L} = \quad & u(c_1) + \beta \sum_s \pi(s) u[c_2(s)] \\ & + \lambda_1 \left[ y_1 - \sum_s q(s)\, b(s) - c_1 \right] + \sum_s \lambda_2(s) \left[ y_2(s) + b(s) - c_2(s) \right]\end{aligned}$$

$\lambda_1, \lambda_2(s)\ s \in S$ are Lagrange multipliers. Under complete markets, we have **one $\lambda_2(s)$ for each state**. F.O.C w.r.t.

$$\begin{aligned} c_1 \quad & u'[c_1] - \lambda_1 = 0 \Leftrightarrow \lambda_1 = u'[c_1] \\ b(s) \quad & -\lambda_1\, q(s) + \lambda_2(s) = 0 \Leftrightarrow \lambda_1\, q(s) = \lambda_2(s) \\ c_2(s) \quad & \beta \pi(s) u'[c_2(s)] - \lambda_2(s) = 0 \Leftrightarrow \lambda_2(s) = \beta \pi(s) u'[c_2(s)] \end{aligned}$$

Combine all three F.O.C., we have

$$u'[c_1] q(s) = \beta\, \pi(s)\, u'[c_2(s)]$$

Alternative derivation by substituting out $c_1, c_2$

# Optimal risk sharing under complete markets

First order conditions give the consumption Euler equation for each state $s \in S$

$$\underbrace{u'[c_1]q(s)}_{\substack{\text{the cost of acquiring the AD for state } s, \\ \text{in terms of period 1 marginal utiltiy}}} = \underbrace{\beta\pi(s)u'[c_2(s)]}_{\substack{\text{discounted benefits from having an} \\ \text{additional unit of consumption} \\ \text{in state } s \text{ in period 2}}}$$

$$\Rightarrow c_1^{-\sigma} q(s) = \beta\pi(s)c_2^{-\sigma}(s) \tag{2}$$

For two different states $s, \tilde{s} \in S$, we have

$$\frac{q(s)}{q(\tilde{s})} = \frac{\pi(s)c_2(s)^{-\sigma}}{\pi(\tilde{s})c_2(\tilde{s})^{-\sigma}}$$

# Optimal risk sharing under complete markets

$$\frac{\pi(s)c_2(s)^{-\sigma}}{\pi(\tilde{s})c_2(\tilde{s})^{-\sigma}} = \frac{q(s)}{q(\tilde{s})} \tag{3}$$

- Interpretation: marginal rate of substitution between consumption in state $s$ and $\tilde{s}$ (LHS) must equal the relative prices (RHS)
- Generally, the country will choose to "tilt" its consumption to states with a low relative AD price, as it is “cheaper” to secure consumption in that state using AD
- Full insurance (= same consumption for all $s$) if

$$\frac{\pi(s)}{\pi(\tilde{s})} = \frac{q(s)}{q(\tilde{s})}$$

Full insurance is possible but not always optimal

# Global equilibrium: considering both countries

World resource constraints

$$c_1 + c_1^* = y_1 + y_1^* \equiv y_1^w$$

$$c_2(s) + c_2^*(s) = y_2(s) + y_2^*(s) \equiv y_2^w(s), \quad \forall s \in S$$

with $y^w$ as the world output.

# Global equilibrium

$$c_1^{-\sigma} q(s) = \beta \pi(s) c_2^{-\sigma}(s)$$

It follows

$$c_2(s) = \left[ \frac{\beta \pi(s)}{q(s)} \right]^{1/\sigma} c_1$$

Analogously, we have for Foreign

$$c_2^*(s) = \left[ \frac{\beta \pi(s)}{q(s)} \right]^{1/\sigma} c_1^*$$

Summing over the countries, we have

$$c_2(s) + c_2^*(s) = \left[ \frac{\beta \pi(s)}{q(s)} \right]^{1/\sigma} (c_1 + c_1^*) \quad \forall s \in S$$

# Global equilibrium (cont.)

$$c_2(s) + c_2^*(s) = \left[\frac{\beta\pi(s)}{q(s)}\right]^{1/\sigma} (c_1 + c_1^*) \quad \forall s \in S$$

Making use of the world resource constraint

$$y_2^w(s) = \left[\frac{\beta\pi(s)}{q(s)}\right]^{1/\sigma} y_1^w \quad \forall s \in S \tag{4}$$

$$\Rightarrow \frac{q(s)}{q(\tilde{s})} = \frac{\pi(s)}{\pi(\tilde{s})} \left[\frac{y_2^w(s)}{y_2^w(\tilde{s})}\right]^{-\sigma} \tag{5}$$

Interpretation: AD for a state $s$ is more expensive if world output $y^w(s)$ is low, or if the probability $\pi(s)$ is high.

# Country vs. global risk

So far, we have derived

$$\frac{q(s)}{q(\tilde{s})} = \frac{\pi(s)}{\pi(\tilde{s})}\left[\frac{y_2^w(s)}{y_2^w(\tilde{s})}\right]^{-\sigma}, \frac{q(s)}{q(\tilde{s})} = \frac{\pi(s)}{\pi(\tilde{s})}\left[\frac{c_2(s)}{c_2(\tilde{s})}\right]^{-\sigma} = \frac{\pi(s)}{\pi(\tilde{s})}\left[\frac{c_2^*(s)}{c_2^*(\tilde{s})}\right]^{-\sigma}$$

$$\Rightarrow \frac{c_2(s)}{c_2(\tilde{s})} = \frac{c_2^*(s)}{c_2^*(\tilde{s})} = \frac{y_2^w(s)}{y_2^w(\tilde{s})}$$

From the consumption Euler equations, we have

$$\left[\frac{\beta\pi(s)}{q(s)}\right]^{1/\sigma} = \frac{c_2(s)}{c_1} = \frac{c_2^*(s)}{c_1^*} = \frac{y_2^W(s)}{y_1^W}$$

$$\Rightarrow \frac{c_1}{y_1^w} = \frac{c_2(s)}{y_2^w(s)} = \frac{c_2(\tilde{s})}{y_2^w(\tilde{s})} \equiv \mu, \qquad \frac{c_1^*}{y_1^w} = \frac{c_2^*(s)}{y_2^w(s)} = \frac{c_2^*(\tilde{s})}{y_2^w(\tilde{s})} \equiv 1-\mu$$

# Optimal international risk sharing

Under complete financial markets:

$$\frac{c_2(s)}{c_1} = \frac{c_2^*(s)}{c_1^*} = \frac{y_2^W(s)}{y_1^W}$$

$$\frac{c_1}{y_1^W} = \frac{c_2(s)}{y_2^W(s)} = \frac{c_2(\tilde{s})}{y_2^W(\tilde{s})} \equiv \mu$$

**Implications**

- Same **consumption growth rates across countries**
- Each country consumes a constant fraction of **world output**

**Economic interpretations:**
With complete international financial markets:

- Countries insure each other against **country-specific shocks**
- As a result, consumption does **not track domestic output**
- Instead, consumption fluctuates with **world output**

# Empirical prediction from optimal international risk sharing

# Do empirical observations support optimal risk sharing?

- The prediction from the international risk sharing is however not reflected in the data
- **Backus-Kehoe-Kydland consumption correlation puzzle:** consumption is much less correlated across countries than output
- **Kose et al. (2009)**: financial globalization has a limited impact on promoting international risk sharing

- Possible explanations due to **(1)** optimal to limit consumption risk sharing or **(2)** sub-optimal risk sharing due to market frictions
  - Home bias in portfolio choices: **(1) and (2)** $\rightarrow$ **Week 3 & Projects**
  - Non-tradable goods **(1)** $\rightarrow$ **Online quiz**
  - Market imperfection or incompleteness **(2)**

## Discuss:

How complete are the international financial markets?

Could you give some example of financial assets that resemble state-contingent assets / non-state-contingent assets?

# Sovereign defaults

Basic concepts and empirical regularities

A simple model of sovereign default

**Discuss:**

# What are sovereign debts?

- Sovereign debt =
  government debt held domestically + **government debt held internationally**
- External debt =
  debt held internationally =
  **government debt held internationally** + private debt held internationally
- Some complications due to different definitions of external debt, which can lead to different level of external debt.
  - Total external debt or government external debt
  - Debt held by foreign residents (banks, firms, households...)
  - **Debt issued in a foreign jurisdiction/foreign currency**, e.g., some Argentinian government bonds were issued in the US, under the US law, and in US dollar

# What count as sovereign defaults?

**Definition of sovereign defaults**

- **Start**: when a country fails to meet a principal or interest payment, as originally stated in the contract, on the due date.
- **End**: when the country resumes payments of interest and principal (including arrears), or after a debt settlement that leads the rating agency to conclude that no further near-term resolution of creditors' claims is likely.

Sovereign defaults can coincide with private sector external default.

# Sovereign default happens frequently: 1800-2014

## Some countries are serial defaulters

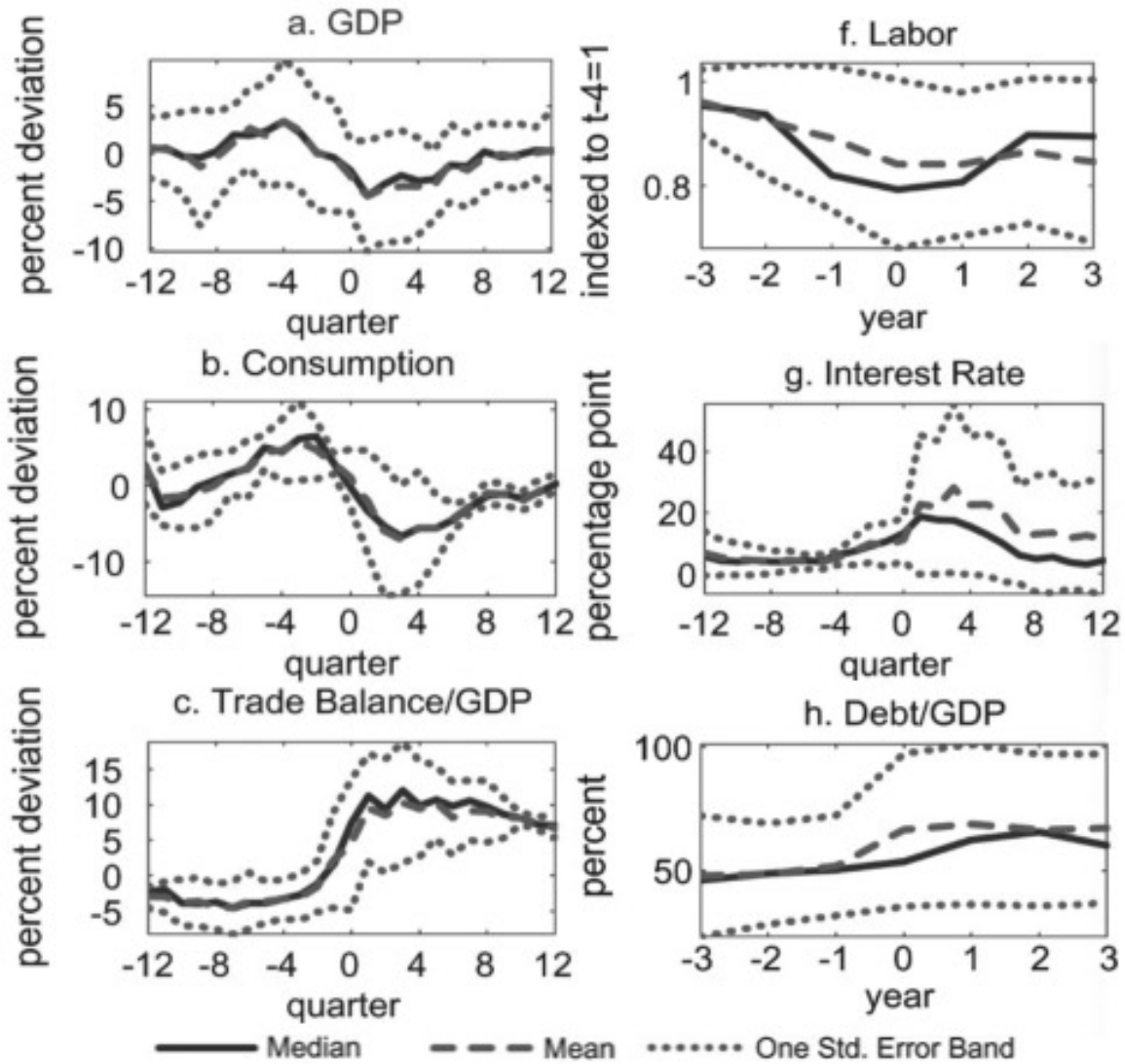
Graphic source: the Economist, Jul 31st 2014, Usual suspects. Data source: Reinhart and Rogoff (2009)

# Large variations in the length of defaults and default probabilities

| Country | Number of Defaults 1824-2014 | Probability of Default | | Years in State of Default per Default Episode |
|---|---|---|---|---|
| | | all years | years not in default | |
| Argentina | 5 | 0.026 | 0.035 | 10 |
| Brazil | 7 | 0.037 | 0.047 | 6 |
| Chile | 3 | 0.016 | 0.020 | 14 |
| Colombia | 7 | 0.037 | 0.058 | 10 |
| Egypt | 2 | 0.010 | 0.012 | 11 |
| Mexico | 8 | 0.042 | 0.056 | 6 |
| Philippines | 1 | 0.005 | 0.006 | 32 |
| Turkey | 6 | 0.031 | 0.037 | 5 |
| Venezuela | 10 | 0.052 | 0.079 | 6 |
| Mean | 5.4 | 0.029 | 0.039 | 11 |

Note. The sample includes only emerging countries with at least one external-debt default or restructuring episode between 1824 and 1999. Source: Uribe and Schmitt-Grohé (2017) Table 13.1

# Business cycle dynamics around sovereign defaults




Source: Mendoza and Yue (2012). A general equilibrium model of sovereign default and business cycles. The Quarterly Journal of Economics, 127(2), 889-946. Fig 1

# How are sovereign defaults resolved?

- IMF and the Paris Club

  Background video: how the IMF facilitate sovereign default restructuring?

- A **haircut** is the investors' losses due to default and restructuring. Empirical studies estimate average of haircuts of 40%, with large variance (13% (Uruguay, 2003) to 73% (Argentina, 2005)).
- The **holdout** problem happens when some creditors refuse to accept the haircut and insist on full repayment instead.
- The **Collective Action Clauses (CACs)** can force minority holdouts to accept the haircut if sufficiently enough investors (supermajority) accept the restructuring terms. CACs can reduce holdouts.

# Why do countries repay their external debts anyway?

- No supranational authority with legal power

  $\rightarrow$ lack of enforceability of international debt contract
- There are costs associated with defaults
- The "willingness-to-pay" matters more than the "ability-to-pay"

  $\rightarrow$ models with endogenous default decisions

**Repayment occurs only if costs exceed benefits of default from the country's perspective!**

# Sovereign defaults

Basic concepts and empirical regularities

A simple model of sovereign default

# Model setup

We focus on the **intuition**. The **formal derivations** are in the appendix.

**Environment**

- **Home**: small open economy with stochastic output
  - Output: $y_1 = \bar{y}, \quad y_2 = y_2(s)$
  - **Budget constraint**: consumption equals output net of debt repayment and new borrowing
  - **Limited commitment**: the government decides whether to repay in period 2
- **Financial market**: incomplete
  - One-period **non-state-contingent bond** $b$
  - Bond price: $q$

**Key feature:** Because repayment is not enforceable, default is a strategic choice.

# Sovereign decision in period 2: costs vs benefits

| **Repay** | **Default** |
|---|---|
| Repay outstanding debt (lower current consumption) | No repayment (higher current consumption) |
| Keep access to international financial markets | Lose market access |
| Avoid output losses | Output loss $\kappa$ |
| Maintain reputation with investors | Reputation loss / financial exclusion |

**Default decision:**
A country defaults if

$$V^{default} > V^{repay}$$

- $V^{default} - V^{repay}$ increases with the utility gain from higher **current consumption**
- decreases with **output costs** of default
- decreases with the utility cost of **exclusion from financial markets**

# Pricing government debt with default risk

In equilibrium, investors must be indifferent between holding risky government debt and a risk-free asset:

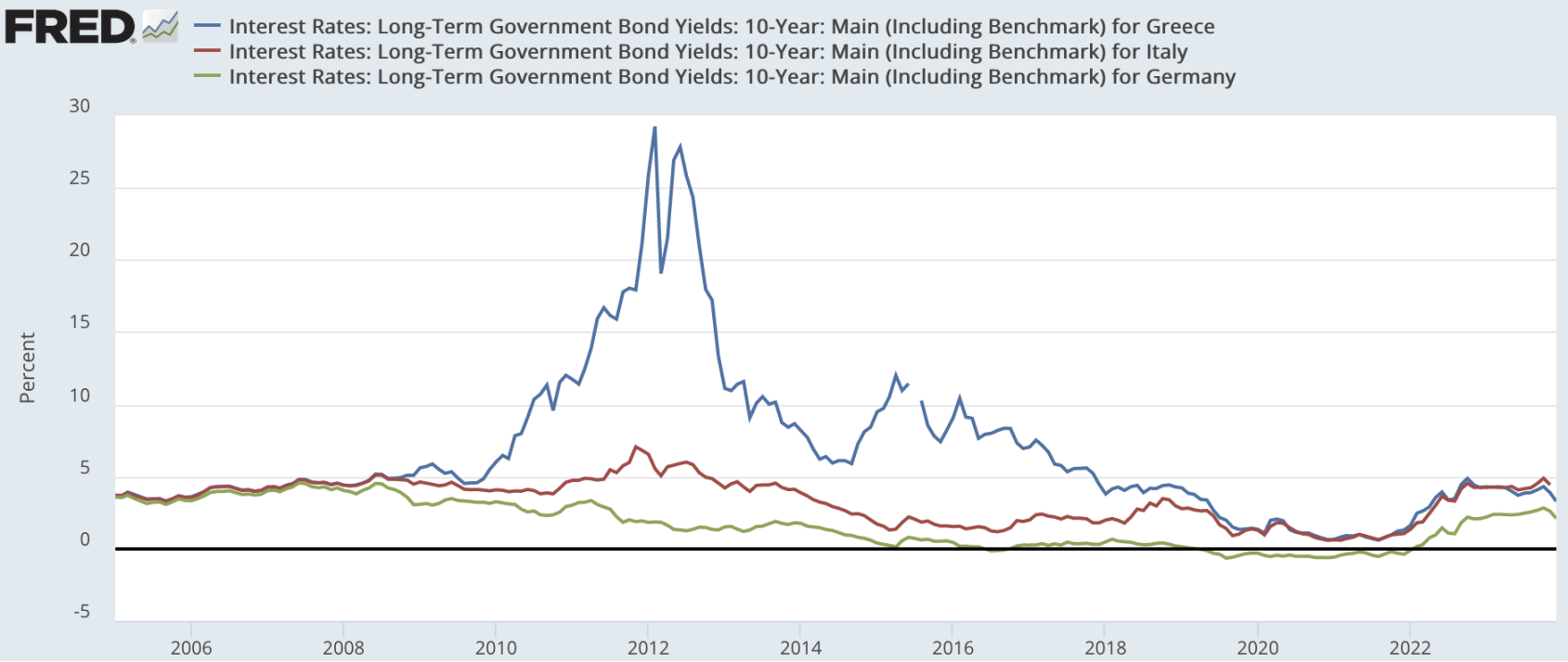
$$q_1 = \frac{1 - prob_D}{R}$$

where $q_1$ : bond price, $prob_D$ : probability of default, $R$ : risk-free return

**Implication**

- Higher default probability $\Rightarrow$ lower bond price
- Lower bond price $\Rightarrow$ higher bond yield
- Investors demand a **spread** to compensate for default risk

# Pricing government debt with default risk: euro area sovereign debt crisis




Source: Organization for Economic Co-operation and Development, Interest Rates: Long-Term Government Bond Yields: 10-Year: for Greece, Italy, and Germany, retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/, March 8, 2024.

# When is default more attractive?

Recall: a country defaults if

$$V^{default} > V^{repay}$$

**Key mechanism:** The benefit of default is avoiding debt repayment, which increases **current consumption**.

**Discussion**
How do the following affect the increase in **current consumption** and thus incentive to default?

- Realized output
- Inherited debt

# When do countries default? The data

## Defaults happen mostly in "bad" times, but sometimes also in "good" times

**The relationship between output and default**

| | Historical data |
|---|---|
| Mean deviation from trend (%) | |
| In first year of default | −1.6 |
| In periods of default | −1.4 |
| In last year of default | −1.3 |
| In periods of non-default | 0.2 |
| Country years below trend (%) | |
| In first year of default | 61.5 |
| In periods of default | 56.2 |
| In last year of default | 58.8 |
| In periods of non-default | 47.2 |

Notes: the sample period 1820 – 2004. Source: **Tomz and Wright (2007) Table 1**

$\rightarrow$ There are other models of sovereign defaults, e.g., due to political turnoever

# When is default more attractive? II

Recall: a country defaults if

$$V^{default} > V^{repay}$$

**Another key mechanism:** The cost of default is smaller when the **output cost is small**

# The incentive to repay depends on the costs of default

- Historically, use of force. This is not uncommon until the beginning of 20th century (Drago doctrine,1907)
- Reputation costs: a defaulter can be excluded from the international financial markets. Financial autarky $\rightarrow$ elevated consumption volatility
- Output losses
  - Foreign investments
  - Productivity
  - Banking sector
  - Export
- Political costs

# Empirical evidence on the output costs of default

**Difficult** to causally identify the output cost of default → discussion later today.

- The existing literature finds negligible to sizable costs

**Table 1**
Existing estimates of the cost of sovereign default.

| Paper | Default cost, % GDP | | Method |
|---|---|---|---|
| | First year | Medium term | |
| *Historical unconditional estimates:* | | | |
| Reinhart and Rogoff (2011a) | 3–4%[a] | 5%[a] | Average path of GDP |
| Tomz and Wright (2007) | 1.6% | 1.4% | Deviation from HP trend |
| *Conditional estimates using more recent data:* | | | |
| De Paoli et al. (2009) | 5.5–10.5%[b] | per year | Fixed effects panel + counterfactual comparison. Defaults with high arrears only. |
| Furceri and Zdzienicka (2012) | 5.6% | 10% | Two-stage GMM panel. Also local projections. Sovereign crises only. |
| Borensztein and Panizza (2008) | 2.6% | not sig. | Fixed effects panel + controls |
| Levy-Yeyati and Panizza (2011) | not sig. | not sig. | Fixed effects panel, quarterly data |

*Notes:* Not sig. means "not significant". All estimates are based on annual data unless otherwise specified.

Source: Kuvshinov and Zimmermann (2019).

- Kuvshinov and Zimmermann (2019): long-lasting output cost of 2.7% of GDP
- Hébert and Schreger (2017): a 10% increase in the probability of default causes a 6% decline in the value of Argentine equities → an especially smart methodology to deal with the **difficulty** of causal identification

# Identifying the empirical output costs of default

**Endogeneity problem** (simultaneity / reverse causality):

Using the data, we can estimate the **correlation** between output drop and sovereign default: do they tend to happen at the same time?

However, this correlation can be due to

- default causes low output, or
- low output trigger default

A potential solution: exogenous trigger of default

Examples:

- Argentina court case in the US (see Hébert and Schreger, 2017)
- Any other suggestions?

# Appendix

# Optimal risk sharing under complete markets: alternative derivation

The maximization problem, where $c_1, c_2$ are substituted out using the budget constraints

$$\max_{\{b(s)\}_S} u\big[\underbrace{y_1 - \sum_s q(s)\, b(s)}_{c_1}\big] + \beta \sum_s \pi(s) u\big[\underbrace{y_2(s) + b(s)}_{c_2(s)}\big]$$

F.O.C. w.r.t. $b(s), \forall s \in S$

$$u'\big[\underbrace{y_1 - \sum_s q(s)\, b(s)}_{c_1}\big](-q(s)) + \beta\, \pi(s)\, u'\big[\underbrace{y_2(s) + b(s)}_{c_2(s)}\big] = 0$$

$$\Rightarrow u'[c_1]q(s) = \beta\, \pi(s)\, u'[c_2(s)]$$

back

# A default model with limited commitment and incomplete markets

- **Home**: a small open economy
  - Period 1: Fixed output of $y_1 = \bar{y}$
  - Period 2: Exogenous output $y_2 = y_2(s)$
  - Period 3: Fixed output of $y_3 = \bar{y}$
  - **limited commitment**: The country *decides* whether to honor the financial contracts in period 2. If not, 100% haircut
  - A linear preference $u(c) = c$
- **Foreign**: competitive international financial markets with a mass of *risk-neutral* investors
- **Incomplete markets**: there is only one-period non-state contingent bond $b$, traded at the price $q$ in period 1 and 2

# Budget constraints

The country's budget constraints

| | |
|---|---|
| **Period 1:** | $c_1 = y_1 - q_1\, b_1, \quad b < 0$ if borrowing |
| **Period 2:** | |
| default | $c_2(s) = y_2(s) - y_2(s)\kappa = y_2(s)(1-\kappa)$ |
| | $y(s)\kappa \geq 0$ : output costs of default |
| repay | the country can buy/issue new debt $b_2$ at a price $q_2$, with $b_2 \geq \bar{b}$, $\bar{b} \in [-\bar{y}, 0]$ |
| | $c_2(s) = y_2(s) + b_1 - q_2\, b_2,$ |
| **Period 3:** | |
| defaulted in period 2 | $c_3 = \bar{y}(1-\kappa)$ |
| repaid in period 2 | $c_3 = \bar{y} + b_2$ |

# Debt decision

## Period 2 and 3

In **period 3:** no default decision in our simple model, debt $-b_2$ is risk free. hence $q_2 = \frac{1}{R}$, where $R$ is the risk-free rate. We assume that the country is impatient

$$\beta < q_2$$

In **period 2:**

- If the country has **defaulted**, it loses access to international financial markets in period 2 and 3. It consumes its output - default costs
- If the country has chosen to **repay** its debt $-b_1$, it chooses $b_2$ to maximizes its expected discounted utility

$$V_{repay}(b_1, s) = \max_{b_2} \left\{ \underbrace{y_2(s) + b_1 - q_2 b_2}_{c_2} + \beta \underbrace{(\bar{y} + b_2)}_{c_3} \right\}$$

subject to $b_2 \geq \bar{b}$.
Under the assumption $q_2 > \beta$, the country will borrow up to the borrowing limit $b_2 = \bar{b} < 0$

# Default risk and bond price

Value of defaulting in period 2: $V_{default}(s) = \underbrace{y_2(s)(1-\kappa)}_{c_2} + \beta \underbrace{\bar{y}\,(1-\kappa)}_{c_3}$

Value of repaying in period 2: $V_{repay}(b_1, s) = \left\{ \underbrace{y_2(s) + b_1 - q_2\bar{b}}_{c_2} + \beta \underbrace{(\bar{y} + \bar{b})}_{c_3} \right\}$

In **period 2**, the country chooses to default if

$$V_{default}(s) > V_{repay}(b_1, s)$$

Probability of default: $prob_D \equiv prob\{s : V_{default}(s) > V_{repay}(\mathbf{b_1}, s)\}$

And $prob_D$ depends on the country's chosen borrowing in period 1 $\mathbf{b_1}$

# Default risk and bond price

In equilibrium:

$$\underbrace{\frac{\text{expected repayment}}{q_1}}_{\text{Expected return on gov. debt with default risk}} = \underbrace{R}_{\text{Return on a risk-free debt}}$$

Thus, equilibrium price $q_1$ for the gov. debt:

$$\begin{aligned} q_1 &= \frac{\text{expected repayment}}{R} \\ &= \frac{(1 - prob_D) * 1 + prob_D * 0}{R} = \frac{1 - prob_D}{R} \\ \text{Bond yield} &= \frac{1}{q_1} = \frac{R}{1 - prob_D} \end{aligned}$$

- Bond price $q_1$ is low (= bond yield is high) when default probability is high
- non-negative spread: $1/q_1 - R \geq 0$

# Debt decision

## Period 1

The country chooses $b_1$ to maximizes expected discounted utility

$$\max_{b_1} \left\{ \underbrace{\bar{y} - q(b_1)\, b_1}_{c_1} + \beta \underbrace{\sum_{s \in S} \pi(s) \max\left[V_{repay}(b_1, s),\, V_{default}(s)\right]}_{\mathbb{E}_1 \max[V_{repay}(b_1), V_{default}]} \right\}$$

Because of the operator $\mathbb{E}_1 \max$, we can't simply take first order conditions. Instead, endogenous default model is generally solved using other numerical methods.

# When do countries default?

**In period 2**, the country chooses to default whenever

$$V_{default}(s) > V_{repay}(b_1, s)$$

$$y_2(s)(1-\kappa) + \beta\bar{y}\,(1-\kappa) > y_2(s) + b_1 - q_2\bar{b} + \beta\left(\bar{y} + \bar{b}\right)$$

$$\Leftrightarrow y_2(s) < \frac{-b_1 + (q_2 - \beta)\bar{b}}{\kappa} - \beta\bar{y}$$

# When do countries default?

**In period 2**, the country chooses to default whenever

$$y_2(s) < \frac{-b_1 + (q_2 - \beta)\bar{b}}{\kappa} - \beta\bar{y}$$

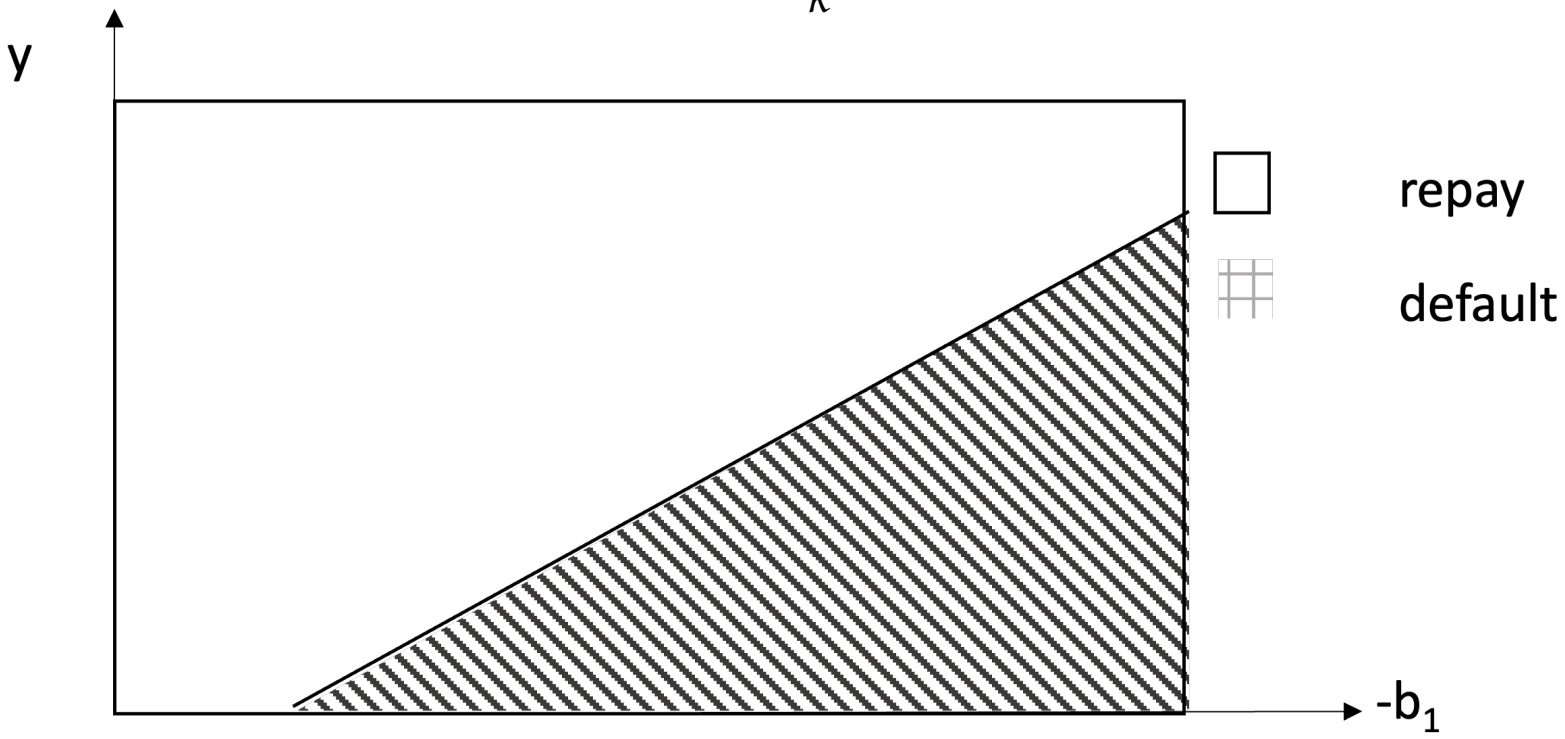
All else being equal, the country defaults when...

- output $y_2(s)$ is low.
- the initial debt $-b_1$ is high
- default costs $\kappa$ are low
- it is difficult to rollover debt
  - bond price $q_2$ to rollover debt is low (e.g. due to increased $R$)
  - borrowing constraint is tight ($\bar{b}$ is high)

# When do countries default? An illustration

There is a state of the world $y^* = y(s^*)$ such that the country is indifferent between defaulting or repaying $V_{repay} = V_{default}$

$$y^* = \frac{-b_1 + (q_2 - \beta)\bar{b}}{\kappa} - \beta\bar{y}$$




**Output in default:** $y_2(s)(1-\kappa)$

# References and further reading I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